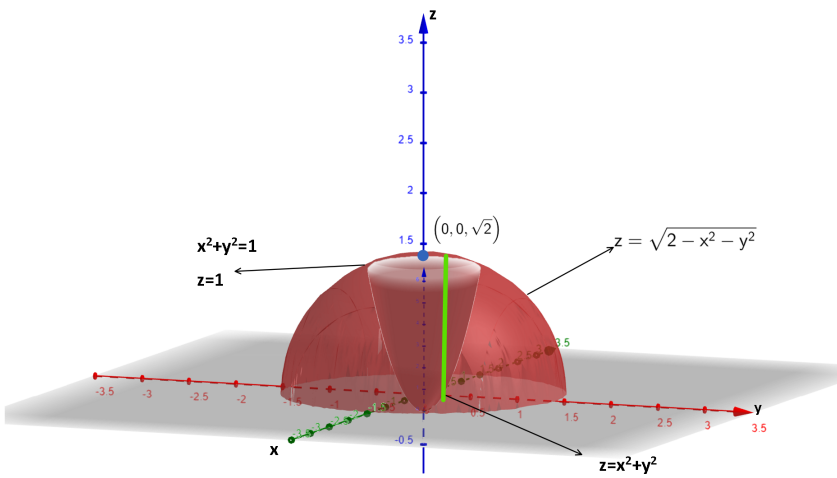
MAST20009 Vector Calculus

Problem Sheet 4: Double and Triple Integrals

Worked Solutions for selected questions

Q80 (a) Volume inside

$$z = x^2 + y^2, \text{ and } \quad z = \sqrt{2 - x^2 - y^2}$$



Solution:
For intersection:

$$z = x^2 + y^2 = \sqrt{2 - x^2 - y^2}$$
$$\Rightarrow z = \sqrt{2 - z^2} \Rightarrow z^2 = 2 - z^2 \Rightarrow z = \pm 1$$

As $z \geq 0$ then $z = 1$ and $x^2 + y^2 = 1$.
Use cylindrical coordinates:

$$x = \rho \cos\phi, \quad y = \rho \sin\phi, \quad \text{Jacobian} = \rho$$

$$0 \leq \rho \leq 1, \quad 0 \leq \phi \leq 2\pi,$$
$$x^2 + y^2 \leq z \leq \sqrt{2 - x^2 - y^2} \Rightarrow \rho^2 \leq z \leq \sqrt{2 - \rho^2}$$

$$\text{Volume} = \int_0^1 \int_0^{2\pi} \int_{\rho^2}^{\sqrt{2-\rho^2}} 1.\rho.\, dz\, d\phi\, d\rho = \int_0^1 \int_0^{2\pi} [\rho z]_{z=\rho^2}^{z=\sqrt{2-\rho^2}}\, d\phi\, d\rho$$
$$= 2\pi \int_0^1 \rho\sqrt{2-\rho^2} - \rho^3\, d\rho = 2\pi \int_0^1 \rho\sqrt{2-\rho^2}\, d\rho - 2\pi \int_0^1 \rho^3\, d\rho$$

For the first integral: Let $u = 2 - \rho^2 \Rightarrow \frac{du}{d\rho} = -2\rho$

$$\int_0^1 \rho\sqrt{2-\rho^2}\, d\rho = \int_2^1 -\frac{1}{2}\sqrt{u}\, du = -\frac{1}{2}.\frac{2}{3}\left[u^{\frac{3}{2}}\right]_{u=2}^{u=1} = \frac{1}{3}\left(2\sqrt{2} - 1\right)$$

For the second integral:

$$\int_0^1 \rho^3\, d\rho = \left[\frac{1}{4}\rho^4\right]_{\rho=0}^{\rho=1} = \frac{1}{4}$$

$$\Rightarrow \text{Volume} = 2\pi\left[\frac{1}{3}\left(2\sqrt{2}-1\right) - \frac{1}{4}\right] = \frac{4\pi\sqrt{2}}{3} + 2\pi\left(\frac{-4-3}{12}\right) = \frac{4\pi\sqrt{2}}{3} - \frac{7\pi}{6}\,.$$

Q83

Solution: Moment of inertia I_y for $x^2+y^2+z^2 \le R^2$, if mass density is ρ

$$I_y = \iiint_D \left(x^2+z^2\right)\rho\, dV$$

Let

$$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$$

$$0 \le r \le R, \quad 0 \le \phi \le 2\pi, \quad 0 \le \theta \le \pi, \quad \text{Jacobian} = r^2\sin\theta$$

$$\begin{aligned} I_y &= \rho\int_0^R\int_0^{2\pi}\int_0^{\pi}\left(r^2\sin^2\theta\cos^2\phi + r^2\cos^2\theta\right) r^2\sin\theta\, d\theta\, d\phi\, dr \\ &= \rho\int_0^R\int_0^{2\pi}\int_0^{\pi} r^4\sin^3\theta\cos^2\phi + r^4\sin\theta\cos^2\theta\, d\theta\, d\phi\, dr \end{aligned}$$

Now using the substitution $u = \cos\theta$ we get

$$\int \sin^3\theta\, d\theta = \int \sin\theta(1-\cos^2\theta)\, d\theta = \int(-1+u^2)\, du$$

$$= -u + \frac{1}{3}u^3 + C = -\cos\theta + \frac{1}{3}\cos^3\theta + C$$

$$\int \sin\theta\cos^2\theta\, d\theta = \int -u^2\, du = -\frac{1}{3}u^3 + C = -\frac{1}{3}\cos^3\theta + C$$

So

$$\begin{aligned} I_y &= \rho\int_0^R\int_0^{2\pi}\left[r^4\cos^2\phi\left(-\cos\theta + \frac{1}{3}\cos^3\theta\right) + r^4\left(-\frac{1}{3}\cos^3\theta\right)\right]_{\theta=0}^{\theta=\pi} d\phi\, dr \\ &= \rho\int_0^R\int_0^{2\pi}\left[r^4\cos^2\phi\left(1-\frac{1}{3}\right) + r^4\left(+\frac{1}{3}\right) - r^4\cos^2\phi\left(-1+\frac{1}{3}\right) - r^4\left(-\frac{1}{3}\right)\right] d\phi\, dr \\ &= \rho\int_0^R\int_0^{2\pi}\left[\frac{4}{3}r^4\cos^2\phi + \frac{2}{3}r^4\right] d\phi\, dr = \rho\int_0^R\int_0^{2\pi}\frac{2}{3}r^4(1+\cos(2\phi)) + \frac{2}{3}r^4\, d\phi\, dr \\ &= \rho\int_0^R\left[\frac{2}{3}r^4(\phi + \frac{\sin(2\phi)}{2}) + \frac{2}{3}r^4\phi\right]_{\phi=0}^{\phi=2\pi} dr \\ &= \rho\int_0^R \frac{4}{3}\pi r^4 + \frac{4}{3}\pi r^4\, dr = \frac{8}{3}\pi\rho\int_0^R r^4\, dr = \frac{8}{3}\pi\rho\left[\frac{r^5}{5}\right]_{r=0}^{r=R} = \frac{8}{15}\pi\rho R^5\,. \end{aligned}$$